吕欣遥

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教育背景

- **同济大学** 上海

  复合型创新人才实验班（数智技术方向） GPA: 4.89/5.00 | 排名: 1/19 2025.03 – 2026.07

  城乡规划学 GPA: 4.89/5.00 | 排名: 2/45 2023.09 – 至今

  主修课程:

  设计课程与空间设计: 设计基础 I、II、III；数智空间设计（一）空间几何图解设计；数智空间设计（二）多模态空间建构；数智空间设计（三）城市系统建模；数智空间设计（四）计算性城市设计（所有课程均为优秀）

  量化与计算技能: 高等数学；概率论与数理统计；Python 程序设计；计算机图形学；基于 Python 的空间分析与建模；人工智能辅助城市设计；城市分析方法与工具；人因空间数据分析（所有课程均为优秀）

专业技能

- **编程**: Python, JavaScript, SQL, React, Node.js
- **设计**: AutoCAD, SketchUp, Rhino, Grasshopper, Blender, Figma, Photoshop, Illustrator, InDesign, V-Ray, D5, Procreate
- **综合素养**: 团队领导力, 活动策划与统筹, 演讲, 学术写作, 管理能力

工作与实习经历

- **互思唯尔（上海）商务咨询工作室** 远程实习

  新媒体运营实习生 *2025.09 – 至今*

  - **内容策划与产出**: 负责微信公众号的内容策划与制作，包括选题研究、文案撰写以及排版等。

- **四川精兴摩托车制造有限公司** 四川

  数据分析实习生 *2025.07 – 2025.08*

  - **用户行为分析**: 对客户数据进行多维度挖掘与分析，识别用户行为模式，为商业决策提供量化支持。
  - **数据清洗与可视化沟通**: 负责大量数据的获取、清洗、预处理和校验，并将处理后的数据转化为直观的数据看板和分析报告，支持团队内部高效沟通。

项目经历

科研项目

- **HCI+ 2026 暑期科研项目** 2026.06 – 至今

  入选学员 *(录取率: 11.6%)* 全球学者群

  在 Prof. Feilin Han 的指导下开展人机交互前沿研究。聚焦于跨文化传播中的感知—认知驱动视频生成，基于眼动追踪与 fNIRS 实验，分析不同文化背景用户对非遗视频内容的视觉注意与认知反应，构建多模态感知数据驱动的设计空间，并结合生成式 AI 实现视频内容重编辑，提升中国文化内容的全球化、精准化传播效果。

- **DigitalFUTURES 2026 国际工作坊 — 人机协作机制** 2026.06 – 2026.07

  核心成员 *Aalto + MIT + GT*

  探讨从预测模型到可解释性人机协同智能的范式转变（指导老师：Prof. P. Fricker, C. Yao, C. Zhu, Y. Yao）。参与开发了基于芝加哥城市数据的智能数字孪生平台，量化分析城市分区、空间特征与犯罪时空分布的关联。利用生成式设计与代理模型，实现了可实时调参的社区形态生成，以辅助城市政策优化。同时基于 GAMA 平台完成了人因行为智能体模拟，并参与部署了一款跨平台移动应用，为居民和游客提供实时安全路线导航。

- **视觉元素深度分布与行人步行行为机制研究** 2025.11 – 2026.02

  核心研究成员 *(审稿完成)* 人因研究

  参与构建了基于深度感知的“空间深度分区–行人关联模型”的理论框架与方法论基础。完成了跨越计算机视觉、空间视觉指标和行人步行机制的跨学科文献综述，为后续的 SHAP 可解释性分析和具体空间设计策略的提出提供了核心支撑。

- **多模态大语言模型与人类交互系统综述研究** 2025.08 – 2025.09
  研究助理 *HCI 研究*
  系统性梳理了过去五年间多模态大语言模型及其与人类行为机制交互的研究成果。构建文献筛选与分类体系，并开发了辅助分析工具以实现高效的文献归纳与整合。深度参与了论文整体理论架构的搭建，并独立绘制了系统概念框架图以清晰呈现核心逻辑。

- **街道立面设计与行人心理反馈机制研究** 2025.07 – 2025.12
  核心负责人 神经建筑学研究
  主导该跨学科研究的项目统筹与实验设计，构建基于 fNIRS 脑成像数据的空间策略模型。设计了完整的实验范式与认知追踪方案，在 AIGC-fNIRS 框架下监测行人在不同模拟街景刺激下的实时心理与认知情绪反馈。建立了立面特征与人类认知之间的关联，将脑电认知数据转化为可直接指导城市更新、数据驱动的空间设计指标。

- **空间金融化国际比较研究** 2025.02 – 2025.07
  课题组核心成员 空间金融化研究
  独立负责日本空间金融化演进路径的案例研究。聚焦过去几十年东京房地产市场的历史演变，以六本木项目作为典型范式，系统梳理并分析了全球资本流入与本地城市空间结构重组之间的深层关联机制。

- **企业迁移与产业动态演化研究** 2024.11
  研究助理 演化经济学研究
  探究了疫情语境下，企业迁移模式与产业动态演化之间的关联特征。利用区域货运数据作为代理指标，观察宏观经济流向并识别产业发展趋势，研究了外部重大冲击如何重新塑造区域产业的时空分布。

应用与开发项目

- **芝加哥城市犯罪模拟系统 (Web 与 Android 应用)**: 直面高犯罪率这一迫切的城市公共安全痛点。项目基于数字孪生技术开发了跨平台应用，旨在通过技术介入为多方提供安全赋能：一方面为居民和游客提供实时的犯罪风险热力图及安全出行路线推荐；另一方面为政府提供政策优化的效益模拟依据，通过评估城市分区变更对安全的潜在影响，探索以环境设计预防犯罪的数智化路径。
- **Living Jade (Android 应用)**: 聚焦于现代社会因快节奏生活导致的精神焦虑、文化认同淡化以及深层社交断裂问题。项目从情感疗愈与文化重构的视角出发，将科学的人格测评与传统玉石文化结合。通过全栈开发，部署了基于大语言模型的智能 AI 玉石专家，以个性化的多模态交互为用户构建兼具情感陪伴、深度对话与文化共鸣的数字场域，缓解个体社交孤立，重塑文化归属感。
- **建筑平面图智能生成工具**: 针对传统建筑设计中长期存在的人工试错成本高、机械性重复劳动繁重这一工作方法的沉疴。工具旨在通过人机协同范式打破设计效率瓶颈，开发了一款能够基于现实世界复杂规划约束条件的生成式计算工具。系统通过自动化求解空间最优布局，将建筑师从底层的规则对齐工作中解放出来，从而有更多精力关注真正具有社会价值的空间人性化设计。

荣誉与奖项

- 连续两年获得国家奖学金 2023-2025
- 连续两年获得优秀学生荣誉称号 2023-2025
- 社会活动奖学金 2024.12
- 中国国际大学生创新大赛 (2024) 银奖（项目: 智联网社区家） 2024.06
- 中国国际大学生创新大赛 (2024) 银奖（项目: 智缆筑界——云端智能控制的绳驱并联建筑机器人） 2024.06
- 上海市女大学生创新创业大赛（创意组）三等奖 2023.12

志愿活动

- **2025 大都市规划国际咨询会（MPIC）** 上海
  深度参与嘉宾接待与引导、物料发放、会务咨询、秩序维持等工作 *2025.12*

- **The 2nd AI4Cities International Conference & Inaugural AI4Cities Workshop** 上海
  负责组织、接待、引导等工作 *2025.07*

- **计算性设计学术论坛暨中国建筑学会计算性设计专业委员会年会** 上海
  负责会议接待、引导、物料制作及发放等工作 *2025.07*

- **同济大学“交想”支援服务队** 上海
  在杨浦滨江公共服务站提供咨询引导服务，协助站点日常运营，参与社区公共活动支持 *2024.04*

Xinyao Lyu

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Education

- **Tongji University** Shanghai, China

  Interdisciplinary Innovation Honors Program (Digital Intelligence Track) GPA: 4.89/5.00 — Rank: 1/19 Mar 2025 – Jul 2026

  Urban Planning GPA: 4.89/5.00 — Rank: 2/45 Sep 2023 – Present

  Relevant Coursework:

  Design Studio & Spatial Design: Design Fundamentals I, II, III; AI-Spatial Design I: Geometric Representation of Space; AI-Spatial Design II: Multimodal Spatial Construction; AI-Spatial Design III: Urban System Modeling; AI-Spatial Design IV: Computational Urban Design (All graded Excellent)

  Quantitative & Computational Skills: Advanced Mathematics; Probability and Mathematical Statistics; Python Programming; Computer Graphics; Spatial Analysis and Modeling with Python; AI-Assisted Urban Design; Urban Analysis Methods and Tools; Human-Oriented Spatial Data Analytics (All graded Excellent)

Skills

- **Programming**: Python, JavaScript, SQL, React, Node.js
- **Design**: AutoCAD, SketchUp, Rhino, Grasshopper, Blender, Figma, Photoshop, Illustrator, InDesign, V-Ray, D5, Procreate
- **Soft Skills**: Leadership, Event Management, Public Speaking, Academic Writing, Time Management

Work Experience

- **Husiweier (Shanghai) Business Consulting Studio** Remote

  New Media Intern *Sep 2025 – Present*
  - **Content Planning and Production**: Planned and produced content for the official WeChat public account, including topic research, copywriting, and layout coordination.

- **Sichuan Jingxing Motorcycle Co., Ltd.** Sichuan, China

  Data Analysis Intern *Jul 2025 – Aug 2025*
  - **Customer Behavior Analysis**: Conducted multi-dimensional analysis of customer data to identify behavioral patterns and support business decision-making.
  - **Data Analysis and Insight Communication**: Conducted data cleaning, preprocessing, and validation, and translated processed data into visual dashboards and reports to support accurate analysis and effective communication of insights.

Projects

Research Projects

- **HCI+ 2026 Summer Research Program** Jun 2026 – Present

  Selected Scholar (Acceptance Rate: 11.6%) *Global Cohort*

  Conducting cutting-edge research in human-computer interaction under the supervision of Prof. Feilin Han. Focusing on perception- and cognition-driven video generation for cross-cultural communication, the project integrates eye-tracking and fNIRS experiments to investigate users' visual attention and cognitive responses to intangible cultural heritage videos. The resulting multimodal behavioral data are used to construct a design space for AI-assisted video re-editing, enhancing the global and personalized dissemination of Chinese cultural content.

- **DigitalFUTURES 2026 Workshop — Human-AI Teaming** Jun 2026 – Jul 2026

  Invited Research Participant *Aalto Univ. + MIT + GT*

  Investigated the paradigm shift from predictive models to explainable coupled intelligence (Advisors: Prof. P. Fricker, C. Yao, C. Zhu, Y. Yao). Developed an AI-driven digital twin platform evaluating correlations between urban zoning, physical features, and spatio-temporal crime patterns in Chicago. Integrated Generative Design with Surrogate Modeling to enable real-time parameter tuning and neighborhood morphology generation for policy optimization. Implemented Agent-Based Modeling via GAMA to simulate human behaviors, deploying a cross-platform mobile application providing real-time safety routing for residents and tourists.

- **Visual Elements Depth Distribution & Pedestrian Walking Behavior** Nov 2025 – Feb 2026

  Co-worker (Review Complete) *Human-Centric Research*

  Contributed to theoretical framework construction and methodological foundation development for a deep perception-based spatial depth zoning–pedestrian trajectory association model. Conducted interdisciplinary literature synthesis spanning computer vision, spatial visual indicators, and pedestrian behavioral mechanisms. Supported variable logic structuring and indicator system refinement for subsequent SHAP-based interpretability analysis and spatial design strategy formulation.

- **Systematic Review of Multimodal LLMs & Human Interaction** — Aug 2025 – Sep 2025
  Research Assistant — *HCI Project*

  Conducted a systematic review of research published over the past five years on multimodal large language models and their interaction with human behavioral mechanisms. Constructed a structured literature screening and classification framework, and developed an auxiliary literature analysis tool to support efficient synthesis and structured aggregation. Contributed to the overall theoretical architecture of the paper and designed the conceptual framework diagram to clarify logical relationships.

- **Street Façade Design & Pedestrian Psychological Responses** — Jul 2025 – Dec 2025
  Project Coordinator (Submitted) — *Neuro-Architectural Study*

  Led project coordination and interdisciplinary research design for constructing a spatial strategy model based on fNIRS neuroimaging data. Formulated the experimental paradigm and cognitive tracking protocols to monitor pedestrians' real-time psychological responses under varied simulated streetscape stimuli. Successfully mapped the causal relationships between multi-dimensional façade visual features and human emotional restoration, translating cognitive data into actionable, data-driven design metrics for urban renewal.

- **International Comparative Study on Spatial Financialization** — Feb 2025 – Jul 2025
  Collaborative Researcher — *Spatial Financialization Study*

  Contributed to a global comparative research initiative by conducting a case study on Japan's spatial financialization. Focused on the evolution of Tokyo's real estate dynamics, using the Roppongi district to analyze mechanisms linking global capital inflows with urban restructuring.

- **Enterprise Migration & Industrial Dynamic Evolution** — Nov 2024
  Research Assistant — *Evolutionary Economics Study*

  Investigated the correlation between corporate relocation patterns and industrial evolution dynamics under the specific context of the pandemic. Analyzed freight Origin-Destination data to deduce macroscopic economic flows, evaluating how external disruptions reshaped spatial-temporal industry distribution and connectivity.

Development Projects

- **Chicago Crime Sim (Web & Android App)**: Directly addressing the critical urban public safety challenges of high crime rates. Developed a cross-platform application leveraging digital twin technology to empower multiple stakeholders: providing residents and tourists with real-time crime risk heatmaps and intelligent, hazard-avoiding safety routing recommendations, while offering municipal governments simulation evidence for policy optimization by evaluating the safety impacts of zoning changes to explore data-driven pathways for crime prevention through environmental design.
- **Living Jade (Android App)**: Targeting mental anxiety, fading cultural identity, and social isolation in modern fast-paced environments, this full-stack project integrates scientific personality assessments with traditional jade aesthetics. By deploying an LLM-powered AI jade expert, it constructs a digital sanctuary for personalized, multimodal interaction—fostering emotional companionship, deep dialogue, and cultural resonance to mitigate individual isolation and rebuild belonging.
- **Constraint-Based Architectural Floorplan Generation Tool**: Aiming to resolve high manual trial-and-error costs and repetitive labor in traditional workflows, this project introduces a human-AI co-creation paradigm. By developing a generative computing tool that automates the translation of complex zoning and regulatory constraints, the system optimizes spatial layouts and liberates architects from tedious rule alignment, allowing them to redirect cognitive resources toward socially meaningful, human-centric design.

Honors and Awards

- National Scholarship for Undergraduate Students **(Two Consecutive Years)** — 2024 – 2025
- Outstanding Student Award, Tongji University **(Two Consecutive Years)** — 2024 – 2025
- Social Activity Scholarship, Tongji University — 2024
- Silver Award **(Two Projects)**, China International College Students' Innovation Competition — 2024
- Third Prize, Shanghai Women College Students Innovation & Entrepreneurship Competition — 2023

Volunteer Experience

- **2025 Metropolitan Planning International Consultation (MPIC)** — Shanghai, China
  Actively involved in guest reception and guidance, material distribution, information desk support, and on-site coordination — *Dec 2025*
- **The 2nd AI4Cities International Conference & Inaugural AI4Cities Workshop** — Shanghai, China
  Responsible for event organization, guest reception, and participant guidance — *July 2025*
- **Computational Design Academic Forum & Annual Conference of the Computational Design Committee, Architectural Society of China** — Shanghai, China
  Responsible for event organization, guest reception, and participant guidance — *July 2025*
- **"Jiaoxiang" Community Volunteer Program** — Shanghai, China
  Provided visitor consultation and guidance services at the Yangpu Riverside public service station, assisted in daily operations, and supported community engagement activities. — *April 2024*